WORKED SOLUTION

STEP II 2007, Question 2

Pure • Cubic Curves and Area

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q2

The key idea

Everything about a cubic's turning points is packed into its derivative. Because the turning points are at $x = p$ and $x = q$, the derivative must be a multiple of $(x-p)(x-q)$, and comparing coefficients gives b and c at once. After that, the cleanest way to get $m - n$ is to integrate the derivative between the turning points, and the area becomes easy once we notice that $y - n$ has a *double* root at $x = q$.

Solution

Part (i)

Differentiating,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 2bx + c.$$

This quadratic vanishes at the turning points $x = p$ and $x = q$, and its leading coefficient is 6, so

$$6x^2 - 2bx + c = 6(x-p)(x-q) = 6x^2 - 6(p+q)x + 6pq.$$

Comparing coefficients,

$$b = 3(p+q), \qquad c = 6pq.$$

Two facts we shall use later. First, since the coefficient of x^3 is positive, $\mathrm{d}y/\mathrm{d}x = 6(x-p)(x-q)$ is positive outside the roots and negative between them, so the curve rises, falls, then rises: the maximum comes first, and $p < q$.

Second, the condition $n > 0$ restricts q. Substituting $x = q$,

$$n = 2q^3 - 3(p+q)q^2 + 6pq^2 = q^2(3p - q),$$

so $n > 0$ means $q < 3p$. Altogether $0 < p < q < 3p$.

Part (ii)

The curve passes through the origin, and

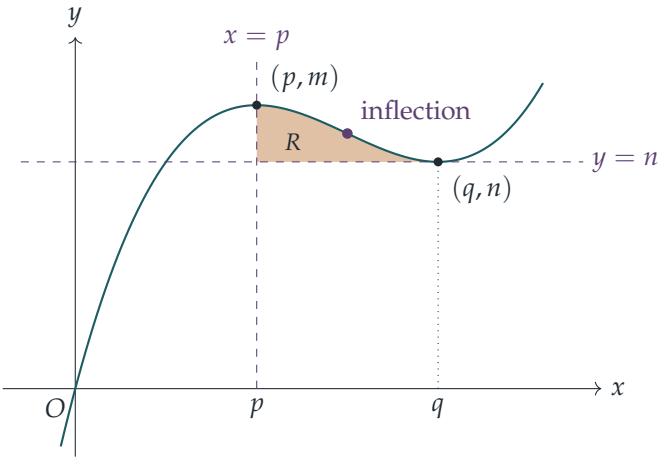
$$y = x(2x^2 - bx + c),$$

where the quadratic factor has discriminant

$$b^2 - 8c = 9(p+q)^2 - 48pq = 3(3p - q)(p - 3q).$$

Since $q < 3p$ and $p < q < 3q$, the first bracket is positive and the second negative, so the discriminant is negative and the origin is the only point where the curve meets the x-axis. So the curve comes up from $-\infty$, crosses the axis at O, rises to the maximum (p, m), falls to the minimum (q, n), still above the axis, and then rises for ever.

The point of inflection is where $\mathrm{d}^2y/\mathrm{d}x^2 = 12x - 2b = 0$, that is, at $x = \frac{b}{6} = \frac{1}{2}(p+q)$, halfway between the turning points.



The region R (shaded) lies between $x = p$ and $x = q$, below the curve and above the line $y = n$; the curve touches $y = n$ at the minimum.

Symmetry. Let $s = \frac{1}{2}(p+q)$, $h = q - p$, and measure from the inflection point by putting $x = s + t$. Then $x - p = t + \frac{1}{2}h$ and $x - q = t - \frac{1}{2}h$, so

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 6\left(t^2 - \tfrac{1}{4}h^2\right),$$

which is an even function of t. Its integral from 0, namely $y - y(s) = 2t^3 - \frac{3}{2}h^2t$, is therefore an odd function of t. So $y(s+t) - y(s) = -\big(y(s-t) - y(s)\big)$: the curve has *rotational symmetry of order* 2 *about its point of inflection* (a half-turn about that point maps the curve to itself). In particular the half-turn swaps the two turning points, so the inflection point is the midpoint of (p, m) and (q, n), namely $\left(\frac{1}{2}(p+q), \frac{1}{2}(m+n)\right)$.

Part (iii)

The drop from the maximum to the minimum is the integral of the gradient:

$$m - n = y(p) - y(q) = -\int_p^q 6(x-p)(x-q)\,\mathrm{d}x.$$

Ask yourself Is there a quick way to evaluate $\int_p^q (x-p)(x-q)\,\mathrm{d}x$ without expanding?

Substitute $x = p + u$, so $x - q = u - h$ with $h = q - p$, and u runs from 0 to h:

$$\int_0^h u(u-h)\,\mathrm{d}u = \frac{h^3}{3} - \frac{h^3}{2} = -\frac{h^3}{6}.$$

Hence

$$m - n = -6\Big(-\frac{(q-p)^3}{6}\Big) = (q-p)^3,$$

as required.

Part (iv)

On $p \le x \le q$ the curve is decreasing from m down to n, so it lies on or above $y = n$, and

$$\text{area of } R = \int_p^q (y-n)\,\mathrm{d}x.$$

The cubic $y - n = 2x^3 - bx^2 + cx - n$ vanishes at $x = q$ and so does its derivative, so $x = q$ is a double root and

$$y - n = 2(x-q)^2(x-r)$$

for some r. Comparing the x^2 coefficients, $-2(2q + r) = -b = -3(p + q)$, so $r = \frac{1}{2}(3p - q)$. With $h = q - p$ again and $u = x - q$ (running from $-h$ to 0),

$$x - r = u + q - \tfrac{1}{2}(3p - q) = u + \tfrac{3}{2}h.$$

Therefore

$$\begin{aligned}\text{area of } R &= \int_{-h}^0 2u^2\big(u + \tfrac{3}{2}h\big)\,\mathrm{d}u = \Big[\tfrac{1}{2}u^4 + hu^3\Big]_{-h}^0 \\ &= -\big(\tfrac{1}{2}h^4 - h^4\big) = \tfrac{1}{2}h^4 = \tfrac{1}{2}(q-p)^4,\end{aligned}$$

which is the required result.

Checks

Take $p = 1$, $q = 2$ (allowed, since $q < 3p$). Then $b = 9$, $c = 12$, and $y = 2x^3 - 9x^2 + 12x$ has $y' = 6(x-1)(x-2)$, with $m = y(1) = 5$ and $n = y(2) = 4 > 0$. So $m - n = 1 = (q-p)^3$. The inflection point is $(1.5, 4.5)$, the midpoint of the turning points, and a numerical integration of $y - 4$ over $[1, 2]$ gives $0.5000 = \frac{1}{2}(q-p)^4$. We also checked all four results for 2000 random pairs with $0 < p < q < 3p$.

Common slips

- Taking $q < p$: for a positive cubic coefficient the maximum is always to the left of the minimum.
- Integrating y instead of $y - n$, which adds the rectangle of height n under the line $y = n$.
- Describing the symmetry as reflection in the line $x = \frac{1}{2}(p + q)$; a cubic has half-turn symmetry about a point, not mirror symmetry.

Beyond STEP

Every cubic is a half-turn symmetric curve about its inflection point, because shifting x to remove the x^2 term (the "depressed cubic" substitution used in Cardano's formula) leaves an odd function plus a constant. The double-root trick in part (iv) is the start of the theory of tangency and multiplicity of roots, which becomes central in algebraic geometry. Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

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